

AIGR Gene Hypothesis Deep Research — AKR1C4 (P17516)

Focus: Aldose-reductase, prostaglandin-metabolism, ketosteroid-monooxygenase, and bile-acid-transport scope

Focus type: function_assignment **Hypothesis slug:** aldose-prostaglandin-monooxygenase-and-bile-transport-scope **Gene:** AKR1C4 / P17516 (human, NCBITaxon:9606) — 3 α -hydroxysteroid dehydrogenase type I / chlordecone reductase / DD4

Executive Judgment

Verdict: REFUTED / OVER-ANNOTATED.

Human AKR1C4 (P17516; DD4 / chlordecone reductase / type-1 3 α -hydroxysteroid dehydrogenase) does **not** directly and curatably hold aldose reductase activity, prostaglandin metabolism, ketosteroid monooxygenase activity, or bile-acid transmembrane transport / transport-process participation. Each of the four contested functions was evaluated independently, as the seed hypothesis requested, and each fails to withstand scrutiny for a different, mutually reinforcing reason:

- 1. Provenance failure.** Live UniProt (P17516) shows that *all four* contested terms rest only on phylogenetic IBA inference (IBA:GO_Central) or legacy TAS:ProtInc carry-over. In direct contrast, *every* experimentally supported (EXP/IDA) molecular-function term for AKR1C4 is a NAD(P)-dependent hydroxysteroid dehydrogenase / ketosteroid reductase activity. No contested term is backed by a direct AKR1C4 assay.
- 2. Chemistry mismatch (monooxygenase).** GO:0047086 (ketosteroid monooxygenase) requires O₂ + NADPH and performs a Baeyer-Villiger oxygen-insertion (progesterone → testosterone acetate). AKR1C4 has zero O₂-utilizing reactions; all ~30 catalytic activities in UniProt are NAD(P)(H)-dependent hydride-transfer reductions/oxidations — the opposite

chemistry. The IBA source study
([P 21232532](#))

itself documents *reductive* HSD chemistry, not monooxygenation.

3. **Paralog attribution (prostaglandin, bile-acid binding).** Primary literature assigns prostaglandin F synthase activity to AKR1C3, AKR1C1/DD1, and the AKR1A/AKR1B subfamilies, and high-affinity bile-acid (lithocholic acid) binding to AKR1C2/DD2 — never to AKR1C4/DD4 in a direct assay. Systematic PGFS screening explicitly treats AKR1C4 as a distinct, non-PGFS isoform.
4. **Structural impossibility (transport).** AKR1C4 is a soluble cytosolic protein with no transmembrane segment. Transmembrane transporter activity (GO:0015125) is structurally impossible; the historical "bile-acid binder" is the paralog AKR1C2, and free-ligand binding is not transmembrane flux.

The most important caveat is that this evaluation rests on public UniProt/GO provenance and on the primary-literature abstracts retrieved; a small residual possibility of weak, unmeasured in-vitro activity cannot be formally excluded for the aldose and prostaglandin functions (see Knowledge Gaps). However, weak in-vitro activity would not justify curatable core-function GO terms. The recommended curation lead is to **remove** the monooxygenase and bile-acid transport terms and to **demote/remove as non-core** the aldose-reductase, prostaglandin, and bile-acid-binding terms, retaining the experimentally grounded 3 α -HSD / chlordecone-reductase core.

Summary

The seed hypothesis asks whether AKR1C4 "retains" four functions beyond its predominant hydroxysteroid-reduction role — aldose reductase, prostaglandin metabolism, ketosteroid monooxygenase, and bile-acid transport — and instructs the evaluator to treat each independently, to read source studies rather than rely on family-name objections, and to look specifically for direct AKR1C4 substrate assays (including weak measurable activity). Over three iterations we performed exactly this: we retrieved live UniProt GO provenance for P17516, computed pairwise sequence identities against the relevant paralogs and outgroups, and read the primary studies cited as evidence sources

([P 8172617](#))

P 8573067

P 9716498

P 21232532

P 2187532

P 23747692

P 22506594

and the Penning functional-plasticity study

P 10998348

The convergent result is that AKR1C4's experimentally validated function is NAD(P)-dependent 3-/17-/20-ketosteroid reduction and 3 α -/17 β -/20 α -hydroxysteroid oxidation — it is among the most catalytically efficient 3 α -HSDs of the four human AKR1C isoforms — plus its eponymous xenobiotic chlordecone-reductase activity. Each contested function instead traces to a specific artifact: an O₂-requiring monooxygenase term that is chemically incompatible with a hydride-transfer reductase; prostaglandin and bile-acid attributes that belong to nearly identical paralogs (AKR1C1/C2/C3, ~82–84% identical) mis-propagated by phylogenetic IBA; a deep-ancestral aldose-reductase term inherited from the distinct AKR1B subfamily (~50% identical); and a transmembrane-transport term that is structurally impossible for a soluble cytosolic protein and that conflates paralog ligand-binding with membrane flux.

The mechanism of over-annotation is transparent and quantifiable. Because the human AKR1C subfamily is a tight cluster of >80%-identical paralogs, function inferred by phylogeny (IBA) and legacy manual carry-over (TAS:ProtInc) spreads a few genuine paralog activities across the whole subfamily. The seed's own premise — that PTHR11732 leaf PTN002482510 descends from ancestral nodes for aldose reductase, ketosteroid monooxygenase, and prostaglandin/bile-acid binding — is precisely the pathway by which these functions were propagated onto AKR1C4 without any direct assay. Reading the source studies confirms that the source chemistry (reductive HSD) does not match the target term names, and that the direct assays name paralogs, not AKR1C4. We therefore refute the hypothesis for curation purposes while flagging the narrow residual uncertainty around weak, untested in-vitro activities.

Key Findings

Finding 1 — All four contested functions are IBA/legacy only; the experimentally supported core is NAD(P)-dependent hydroxysteroid dehydrogenase

Live UniProt GO annotations for P17516 (fetched 2026) partition cleanly by evidence code. The four contested functions carry only weak evidence:

- Aldose reductase (NADPH) — **GO:0004032**, IBA:GO_Central
- Bile acid binding — **GO:0032052**, IBA
- Bile acid transmembrane transporter activity — **GO:0015125**, TAS:ProtInc (legacy)
- Ketosteroid monooxygenase — **GO:0047086**, IBA
- Prostaglandin metabolic process — **GO:0006693**, IBA
- Bile acid and bile salt transport — **GO:0015721**, TAS:ProtInc (legacy)

By contrast, **every** EXP/IDA-supported molecular-function term is an oxidoreductase acting on steroids: 3 α -HSD [NAD(P)⁺] (GO:0140169, EXP), androsterone dehydrogenase (GO:0047023, IDA), chlordecone reductase (GO:0047743, IDA), estradiol 17 β -dehydrogenase (EXP), testosterone dehydrogenase (EXP), and 5 α -androstane-3 β ,17 β -diol dehydrogenase (EXP). The ~30 catalytic-activity reactions listed in UniProt are all NAD(P)(H)-linked carbonyl reductions or hydroxyl oxidations. Penning's functional-plasticity study ([PMID: 10998348](#)) establishes that all four human AKR1C isoforms "*acted as NAD(P)(H)-dependent 3-, 17- and 20-ketosteroid reductases and as 3 α -, 17 β - and 20 α -hydroxysteroid oxidases*," with AKR1C4 the most catalytically efficient 3 α -HSD (kcat/Km 10–30 $\times$ the others). This finding sets the evidentiary baseline: AKR1C4's curatable function is HSD chemistry, and the contested terms have no direct support.

Finding 2 — Ketosteroid monooxygenase (GO:0047086) is a chemistry mismatch

GO:0047086 requires molecular O₂ plus NADPH and converts progesterone to testosterone acetate via a Baeyer-Villiger oxygen-insertion. AKR1C4 has **zero** O₂-utilizing/monooxygenase reactions in UniProt; all listed activities are NAD(P)(H)-dependent hydride transfers with no O₂ co-substrate. Critically, the IBA source study that traces this term to AKR1C1/C2/C3 ([PMID: 21232532](#)) describes the opposite chemistry: "*progesterone is metabolized by reductive 20 α -hydroxysteroid dehydrogenases (20 α -HSDs), 3 α / β -HSDs and 5 α / β -reductases*" — i.e., carbonyl

reduction (progesterone → 20 α -hydroxyprogesterone), not O₂-dependent monooxygenation. AKR1C4 itself was not assayed in that study. The seed's own observation that "live GO:0047086 requires O₂+NADPH+progesterone to testosterone acetate, not progesterone carbonyl reduction" is thus confirmed: the term name and the source chemistry are incompatible. This is not a family-name objection — it is a mechanistic contradiction. The term should be removed.

Finding 3 — Bile-acid transmembrane transport is structurally impossible; the binder is the paralog AKR1C2/DD2

UniProt localizes AKR1C4 to the cytoplasm/cytosol with zero transmembrane or signal features. A transmembrane transporter (GO:0015125) requires membrane-spanning segments that AKR1C4 does not possess. The molecular cloning study ([PMID: 8172617](#)) states that "*the nucleotide and amino acid sequences of DD2 and DD4 are virtually identical with those of human bile-acid binder and human chlordecone reductase cDNAs respectively*" — assigning the bile-acid-binder identity to **DD2 (AKR1C2)** and identifying **DD4 (AKR1C4)** with chlordecone reductase. The follow-up study ([PMID: 8573067](#)) shows the high-affinity lithocholic-acid binding resides in DD2: "*DD1 ... possesses prostaglandin F synthase activity but low affinity for lithocholic acid, whereas DD2 ... exhibited high-affinity binding for the bile acid.*" This is intracellular **ligand binding**, not transmembrane flux. Reading the full source as the seed requested confirms the seed's own caution — abstract emphasis does not exclude other functions — but the data still place the bile-acid attribute on AKR1C2, and no AKR1C4 transmembrane transport assay exists. The two transport terms (GO:0015125, GO:0015721) are TAS:ProtInc legacy; the binding term (GO:0032052) is IBA. All should be removed or demoted.

Finding 4 — Prostaglandin and aldose activities trace to paralogs/other subfamilies, not to direct AKR1C4 assays

Prostaglandin F synthase / 11-ketoprostaglandin reductase activity is an established function of AKR1C3 (type-5 17 β -HSD/PGF synthase; [PMID: 30012349](#), [PMID: 35489629](#)) and of DD1/AKR1C1 ([PMID: 8573067](#): "*DD1 ... possesses prostaglandin F synthase activity*"). No primary study demonstrates significant PG-synthase activity for AKR1C4/DD4. Aldose reductase is the defining activity of the separate AKR1B subfamily (AKR1B1); the AKR1C4 term GO:0004032 is IBA:GO_Central, inherited from a deep ancestral AKR node (seed: PTN000198921), with no direct AKR1C4 glucose/galactose reduction assay identified, and no aldose (sugar) substrate among AKR1C4's UniProt reactions. This directly addresses the seed's request to distinguish source-mapping chemistry from a family-name objection — and to include AKR1C2 (not just

AKR1C3) in the prostaglandin source: even granting AKR1C2

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9716498 as PG sources, the activity still belongs to paralogs, never to AKR1C4.

Finding 5 — Sequence identity confirms the over-annotation mechanism

We computed the global Needleman-Wunsch pairwise identity of AKR1C4 (P17516) live from UniProt sequences:

Paralog / outgroup	Function	% identity to AKR1C4
AKR1C3	PGF synthase / type-5 17 β -HSD	84.0%
AKR1C1 (DD1)	20 α -HSD / PGF synthase	82.8%
AKR1C2 (DD2)	bile-acid binder	81.6%
AKR1D1	steroid 5 β -reductase	57.4%
AKR1B1	aldose reductase	50.6%
AKR1B10	aldose/retinal reductase	50.3%

The tight AKR1C cluster (>80% identical) is exactly the regime in which IBA propagates a paralog's genuine activity across all members. The true PG-synthase enzyme AKR1C3 is 84% identical, while aldose reductase (AKR1B1) is only ~50% identical — an outgroup. The original chlordecone-reductase cloning paper ([PMID: 2187532](#)) independently reports that CDR shows "65% similarity to the primary structure of human liver aldehyde reductase and 66% similarity to the inferred protein sequence of rat lens aldose reductase" and notes nucleotide similarity to bovine lung prostaglandin F synthase — i.e., the aldose and prostaglandin annotations arose from **sequence homology, not measured AKR1C4 activity**.

Finding 6 — Systematic PGFS profiling never implicates AKR1C4; inhibitor studies treat it as a distinct isoform

The systematic PGFS screen of human and bovine AKRs ([PMID: 23747692](#)) reports: "We identified AKR1A1 and confirmed AKR1B1 as the most potent PGFS." AKR1C4 was not identified as a PGFS. Penning's inhibitor work localizes family PG activity specifically to AKR1C3 (PGH₂ → PGF₂ α ; PGD₂ → 11 β -PGF₂; [PMID: 19010312](#), [PMID: 21087665](#)). The baccharin inhibitor study ([PMID: 22506594](#)) experimentally distinguishes AKR1C3 from "other AKR1C isoforms (AKR1C1,

AKR1C2, and AKR1C4)," showing "no significant inhibition toward" them. Across every comparative and inhibitor study retrieved, AKR1C4 is treated as a separate, non-PGFS isoform, and no direct AKR1C4 PG-synthase or aldose (sugar) reduction activity is reported.

Mechanistic Model / Interpretation

The picture that emerges is an over-annotation pipeline driven by paralog density and legacy carry-over, sitting atop a well-defined true function.

TRUE, EXPERIMENTALLY SUPPORTED CORE

AKR1C4 (P17516) = soluble cytosolic NAD(P)-dependent oxidoreductase

- 3 α -HSD [NAD(P)+] (G0:0140169, EXP) ← most efficient AKR1C 3 α -HSD
- androsterone dehydrogenase (G0:0047023, IDA)
- chlordecone reductase (G0:0047743, IDA) ← eponymous xenobiotic
- 17 β - / 20 α -HSD, testosterone / estradiol DH (EXP)

Chemistry: NAD(P)(H) hydride transfer; carbonyl $\leftrightarrow$ hydroxyl. NO 02.

CONTESTED TERMS – HOW EACH WAS PROPAGATED

G0:0004032 aldose red.	IBA from deep AKR1B ancestor (~50% id). AKR1B1 is the real aldose reductase.
G0:0006693 PG metabolism	IBA from AKR1C1/C2/C3 (~82-84% id). Real PGFS = AKR1C3, AKR1C1, AKR1A1/AKR1B1.
G0:0047086 ketosteroid monooxygenase	IBA from AKR1C1/C2/C3. CHEMISTRY MISMATCH: term needs 02; source does reductive HSD.
G0:0032052 bile acid bind G0:0015125 BA transporter G0:0015721 BA transport	IBA; real binder = AKR1C2/DD2 (binding, TAS:ProtInc legacy; STRUCTURALLY IMPOSSIBLE TAS:ProtInc legacy; no TM segment.

Four distinct artifact classes are at work, and it is worth keeping them separate for curation:

- **Ancestral outgroup inheritance** (aldose reductase): the term rides down from a node predating the AKR1B/AKR1C split; ~50% identity is far below the threshold at which one would transfer a substrate-specific catalytic activity.
- **Tight-paralog spread** (prostaglandin metabolism, bile-acid binding): >80% identity makes IBA transfer superficially reasonable, but direct assays name the paralogs, and comparative/inhibitor studies actively exclude AKR1C4.

- **Ontology/chemistry artifact** (ketosteroid monooxygenase): the term is not just mis-attributed, it names a reaction class (O_2 -dependent oxygen insertion) that a hydride-transfer reductase cannot catalyze; the source study documents reductive chemistry.
- **Legacy manual carry-over into an impossible cellular context** (transmembrane transport): TAS:ProtInc annotations from an era of looser standards, asserting membrane transport for a protein with no membrane domain.

The unifying interpretation: AKR1C4 is a promiscuous but mechanistically constrained cytosolic NAD(P)-dependent carbonyl reductase / hydroxysteroid oxidase. Its genuine breadth (steroids + the xenobiotic chlordane) is real, but that breadth has been over-extended by inference into monooxygenation, sugar reduction, eicosanoid synthesis, and membrane transport — activities that either belong to relatives or are chemically/structurally precluded.

Evidence Base (Evidence Matrix)

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
10998348	Direct assay (recombinant enzymology)	Refutes contested; defines core	Is AKR1C4's core function HSD or monooxygenase/ aldose/ transport?	All 4 AKR1C isoforms are NAD(P)(H)-dependent 3/17/20-ketosteroid reductases & HSD oxidases; AKR1C4 most efficient 3 α -HSD	Human recombinant AKR1C1–C4
UniProt P17516 (GO provenance)	Review/ database	Refutes	Evidence codes behind contested terms	All 4 contested terms are IBA or TAS:ProtInc; all EXP/IDA terms are HSD/ oxidoreductase	Curated database
21232532	Direct assay (cell/enzyme)	Refutes (monooxygenase)	Does the monooxygenase IBA source show O ₂ chemistry?	Source documents <i>reductive</i> 20 α /3 α / β -HSD & 5 α / β -reductase metabolism of progesterone; AKR1C4 not assayed	AKR1C1/C2/ C3, ovarian endometriosis
8172617	Structural/ sequence	Refutes (transport/ binding)	Is AKR1C4 the bile-acid binder?	DD2=bile-acid binder (AKR1C2); DD4=chlordecone reductase (AKR1C4)	Human liver cDNA
8573067	Direct assay (binding + PGFS)	Refutes (binding & PG) / Qualifies	Where do bile-acid binding & PGFS reside?	High-affinity lithocholic-acid binding = DD2; PGF synthase =	Human liver isoforms

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
				DD1; described as binding, not transport	
9716498	Sequence/ expression	Qualifies	AKR1C2 identity & tissue distribution	Confirms AKR1C2/DD2 as distinct principal gene; clarifies paralog cDNA confusion	Human tissues
2187532	Sequence/ cloning	Refutes (aldose/ PG basis)	Origin of aldose/ PG resemblance	CDR ~65–66% similar to aldehyde/aldose reductase; nucleotide similarity to bovine PGF synthase = homology, not assay	Human liver CDR cloning
Pairwise identity (computed)	Computational	Supports mechanism	Quantify paralog vs outgroup distance	AKR1C3 84%, AKR1C1 82.8%, AKR1C2 81.6% vs AKR1B1 50.6%	UniProt sequences, live NW alignment
23747692	Direct assay (screen)	Refutes (PG)	Which AKRs are PGFS?	AKR1A1 identified, AKR1B1 confirmed as most potent PGFS; AKR1C4 not a PGFS	Human & bovine AKRs
22506594	Direct assay (inhibitor)	Refutes (PG)	Is AKR1C4 = the PGFS AKR1C3?	Baccharin inhibits AKR1C3, "no significant	Recombinant enzymes

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
				inhibition toward ... AKR1C4"	
19010312	Direct assay	Qualifies (PG→AKR1C3)	Locus of family PG activity	AKR1C3 reduces PGH ₂ →PGF ₂ α, PGD ₂ →11β-PGF ₂ in breast cells	MCF-7
21087665	Review/ structural	Qualifies (PG→AKR1C3)	PG activity & isoform selectivity	AKR1C3 is the PGFS; distinct binding pockets vs AKR1C1/C2	Structural review
11158055	Direct assay	Qualifies (core)	Type-1 vs type-3 3α-HSD properties	Type-1 (AKR1C4) is stable liver 3α-HSD; efficiently converts DHT→3α-diol	Human HEK-293

GO Curation Implications

The following are **leads requiring curator verification**, organized by term.

GO ID	Term	Aspect	Current evidence	Recommended action
GO:0047086	ketosteroid monooxygenase activity	MF	IBA	Remove. Chemistry mismatch — requires O ₂ ; AKR1C4 has no monooxygenase reactions; source (P 21232532) is reductive.
GO:0015125	bile acid transmembrane transporter activity	MF	TAS:ProtInc (legacy)	Remove. Structurally impossible for a soluble cytosolic protein with no TM segment.
GO:0015721	bile acid and bile salt transport	BP	TAS:ProtInc (legacy)	Remove (or at most non-core). No transmembrane flux assay; binder is AKR1C2.
GO:0032052	bile acid binding	MF	IBA	Demote to non-core / remove. High-affinity binder is AKR1C2/DD2; AKR1C4 not directly shown.
GO:0004032	aldose reductase (NADPH) activity	MF	IBA:GO_Central	Demote to non-core / remove. Inherited from AKR1B outgroup (~50% id); no direct AKR1C4 sugar-reduction assay.
GO:0006693	prostaglandin metabolic process	BP	IBA	Demote to non-core / remove. PGFS activity belongs to AKR1C3/AKR1C1/AKR1A1; AKR1C4 excluded in screens.
GO:0140169	3 α -HSD [NAD(P) ⁺] activity	MF	EXP	Retain (core).
GO:0047023	androsterone dehydrogenase activity	MF	IDA	Retain (core).

GO ID	Term	Aspect	Current evidence	Recommended action
GO:0047743	chlordecone reductase activity	MF	IDA	Retain (core).

Recommendation summary: retain the NAD(P)-dependent hydroxysteroid dehydrogenase / ketosteroid reductase and chlordecone-reductase core; remove the two structurally/chemically impossible terms (monooxygenase, transmembrane transporter); and treat aldose reductase, prostaglandin metabolism, and bile-acid binding/transport as non-core inference to be removed or clearly qualified. We deliberately avoid recommending "protein binding" as a fallback — the informative core (3 α -HSD/chlordecone reductase) is the appropriate anchor.

Mechanistic Scope

The immediate molecular function being tested is single-protein catalytic / transport activity of the AKR1C4 gene product. Our analysis keeps direct gene-product activity separate from downstream or inferred roles:

- **Direct, experimentally established:** NAD(P)(H)-dependent hydride-transfer reduction of 3-/17-/20-ketosteroids and oxidation of 3 α -/17 β -/20 α -hydroxysteroids; reduction of the xenobiotic chlordecone. These are cytosolic, cofactor-linked, single-substrate reactions.
- **Contested, not direct:** monooxygenation (an O₂-dependent oxygen-insertion reaction of a different EC class), aldose (sugar aldehyde) reduction, prostaglandin endoperoxide/ketone reduction, and bile-acid membrane transport. None is demonstrated by a direct AKR1C4 assay; each is either inferred by homology (IBA) or carried over from legacy manual annotation (TAS).

Downstream phenotypes (e.g., roles of the family in steroid-hormone balance, cancer proliferation, or bile-acid physiology) are real for specific paralogs but must not be conflated with AKR1C4's own immediate catalytic scope. In particular, the prostaglandin and bile-acid roles are properties of AKR1C3 and AKR1C2 respectively, and any AKR1C4 attribution is inference, not activity.

Conflicts and Alternatives

- **Paralog confusion is the dominant conflict.** The >80% identity among human AKR1C1–C4 makes cDNA and function assignments error-prone;

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explicitly documents multiple near-identical cDNAs (differing by 1–5 nucleotides) that were historically misassigned across DD1/DD2/DD4. This is the single most important source of the contested annotations.

- **Isoform-specific findings.**

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places PGFS on DD1 and bile-acid binding on DD2;

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excludes AKR1C4 from AKR1C3-type PGFS inhibition. These are the strongest direct contradictions to attributing PG/binding to AKR1C4.

- **The seed's fair caution, addressed.** The seed warns that abstract emphasis

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highlighting DD4=chlordecone reductase, DD2=bile-acid binder) does not by itself exclude other tested functions, and that soluble structure is not proof against every transport-process contribution. We agree in principle. However: (a) transmembrane transporter activity specifically requires a membrane domain AKR1C4 lacks, so GO:0015125 is excluded on structural grounds regardless of abstract emphasis; and (b) no full-text assay showing AKR1C4 bile-acid flux or carrier participation was located. An intracellular carrier role remains formally untested (see Knowledge Gaps) but is not a basis for the current transmembrane-transporter term.

- **Weak in-vitro activity possibility.** The seed asks specifically for weak measurable aldose/PG activity. Given ~50% identity to AKR1B1, trace aldose reductase activity is conceivable but was not found in any retrieved assay; even if present, it would not justify a curatable core MF term.
- **Organism differences.** Rodent 3 α -HSDs (e.g., AKR1C9) and canine AKR1C3 show their own substrate profiles

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cross-species carry-over is another possible route for spurious human AKR1C4 annotations, though the main artifacts here are within-human-paralog and ancestral.

Limitations and Knowledge Gaps

1. **No direct AKR1C4 aldose-reductase assay was located.** *Checked:* UniProt reactions (no sugar substrates), PGFS/AKR screens, cloning papers. *Why it matters:* the seed explicitly requests weak measurable activity; absence of evidence is not evidence of absence. *Resolution:* a purified-enzyme kinetic assay of AKR1C4 against glucose/galactose/glyceraldehyde with NADPH.

2. **No direct AKR1C4 prostaglandin-substrate assay was located.** *Checked:*

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screen,

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21087665/22506594. *Why it matters:* same as above. *Resolution:* PGH₂/PGD₂ reduction assay of purified AKR1C4 with LC-MS product ID.

3. **Intracellular bile-acid carrier participation is untested for AKR1C4.** *Checked:*

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8573067 (binding assigned to AKR1C2). *Why it matters:* transport-process participation (as a cytosolic carrier) differs from transmembrane transporter activity. *Resolution:* direct lithocholate-binding affinity measurement for AKR1C4 and a cytosolic-shuttle functional assay.

4. **Provenance is database-derived (UniProt/GO live fetch), not re-derived from PAINT trees.** *Why it matters:* IBA propagation details (exact ancestral nodes) were taken from the seed and UniProt, not independently reconstructed. *Resolution:* inspect the PTHR11732 PAINT annotation and QuickGO ancestry directly.

5. **Full-text of key historical papers not read line-by-line.** We relied on abstracts and cited snippets. *Why it matters:* the seed asks to read complete source studies. *Resolution:* full-text retrieval of

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Discriminating Tests

1. **Purified-enzyme substrate panel.** Assay recombinant AKR1C4 head-to-head with AKR1B1 (aldose control), AKR1C3 (PGFS control), and AKR1C2 (bile-acid control) against: D-glucose/galactose/DL-glyceraldehyde (aldose), PGH₂/PGD₂ (prostaglandin), and lithocholic acid (binding). Quantify kcat/Km; a >100-fold deficit vs the positive-control paralog would confirm non-core status.
2. **LC-MS product identification** for any borderline PG or aldose turnover to distinguish genuine catalysis from background.
3. **Ontology/chemistry audit** of GO:0047086 mapping onto AKR1C isoforms — verify against QuickGO whether the IDA
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actually supports an O₂-dependent reaction or was mis-mapped from reductive HSD data.
4. **Structure/topology check** (already strongly indicated): confirm absence of TM segments via the AlphaFold model and UniProt features to formally exclude GO:0015125.
5. **PAINT tree inspection** of PTHR11732 to document the exact IBA propagation path for each contested term.

Proposed Follow-up Actions / Curation Leads (require curator verification)

Candidate reference snippets to verify (exact quotes):

- P 10998348
"All enzymes acted as NAD(P)(H)-dependent 3-, 17- and 20-ketosteroid reductases and as 3alpha-, 17beta- and 20alpha-hydroxysteroid oxidases." → anchors the core MF.
- P 21232532
"progesterone is metabolized by reductive 20alpha-hydroxysteroid dehydrogenases (20alpha-HSDs), 3alpha/beta-HSDs and 5alpha/beta-reductases" → refutes monooxygenase term.
- P 8172617
"the nucleotide and amino acid sequences of DD2 and DD4 are virtually identical with those of human bile-acid binder and human chlordecone reductase cDNAs respectively" → assigns binder to AKR1C2, AKR1C4 to CDR.
- P 8573067
"DD1 ... possesses prostaglandin F synthase activity but low affinity for lithocholic acid, whereas DD2 ... exhibited high-affinity binding for the bile acid" → places PG on DD1, binding on DD2.
- P 23747692
"We identified AKR1A1 and confirmed AKR1B1 as the most potent PGFS" → PG activity not in AKR1C4.
- P 22506594
"no significant inhibition toward other AKR1C isoforms (AKR1C1, AKR1C2, and AKR1C4)" → AKR1C4 distinct from PGFS AKR1C3.
- P 2187532
"65% similarity to ... human liver aldehyde reductase and 66% similarity to ... rat lens aldose reductase" → aldose annotation is homology.

Candidate action changes:

- Remove GO:0047086 (ketosteroid monooxygenase) and GO:0015125 (bile acid transmembrane transporter activity) — impossible chemistry / topology.
- Remove or mark non-core GO:0015721 (bile acid transport), GO:0032052 (bile acid binding), GO:0004032 (aldose reductase), GO:0006693 (prostaglandin metabolic process) — paralog/ancestral IBA or legacy TAS with no direct AKR1C4 assay.
- Retain GO:0140169, GO:0047023, GO:0047743 and related HSD terms as the experimentally supported core.

Suggested curator questions:

- Does any full-text assay report weak AKR1C4 aldose or prostaglandin turnover? If yes, curate as non-core in-vitro activity, not core function.
- Should any transport-process participation be re-expressed as intracellular ligand binding rather than transmembrane transporter activity, pending direct data?

Suggested experiments: the purified-enzyme substrate panel and LC-MS product ID described under Discriminating Tests.

Conclusion

Evaluated independently as the seed required, none of the four contested functions of human AKR1C4 (aldose reductase, prostaglandin metabolism, ketosteroid monooxygenase, bile-acid transmembrane transport) survives as a directly supported, curatable function. Each rests on phylogenetic IBA inference or legacy TAS carry-over and is contradicted by chemistry (monooxygenase), structure (transport), or direct paralog attribution (prostaglandin, bile-acid binding). The experimentally grounded identity of AKR1C4 is a soluble cytosolic, highly efficient NAD(P)-dependent 3 α -hydroxysteroid dehydrogenase / ketosteroid reductase that also reduces the xenobiotic chlordecone. The hypothesis is **refuted** / **over-annotated** for curation, with narrow residual uncertainty only around unmeasured weak in-vitro aldose/prostaglandin activity that, even if real, would not warrant core GO terms.